10+ PEPTIDES	MECHANISM GUIDE	DOSING PROTOCOLS	STACKING GUIDE

THE COMPLETE PEPTIDE PROTOCOL BIBLE

BPC-157 · TB-500 · GHK-Cu · Semax · Selank · Epithalon · PT-141 · AOD-9604 · Stacking Guide

Mechanism of action · Clinical dosing · Injection protocols · Full stacking guide

ELITE FITNESS GUIDE SERIES	Research-Based • Professional Edition • 2024

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■ 1. WHAT ARE PEPTIDES? THE SCIENCE

Peptides are short chains of amino acids (2–50 amino acids) that act as **biological signaling molecules**. Unlike anabolic steroids (which force supraphysiological responses), peptides work by **activating, amplifying, or modulating your body's own biological pathways** — often with a significantly more favorable safety profile.

> ■ *The peptide research field exploded after the discovery that specific amino acid sequences could modulate growth factor expression, cellular repair mechanisms, and neurotransmitter function without the systemic side effects of full-length hormones.*

Property	Small Molecule Drugs	Anabolic Steroids	Peptides
Mechanism	Enzyme inhibition / receptor blockade	Nuclear AR activation	Receptor activation / signaling modulation
Specificity	Variable	Systemic (all AR-containing tissues)	Often highly tissue-selective
Safety Profile	Variable	Significant systemic side effects	Generally favorable; short half-lives
Detectability (WADA)	Variable	Long detection windows	Most are short-window or undetectable
Legal Status	RX or OTC	Schedule III (USA) — controlled	Research chemicals (grey area in most countries)

Administration Routes:

- **Subcutaneous (SC):** Injected into fat layer under skin — standard for most peptides (BPC-157, TB-500, GHRP, CJC)
- **Intramuscular (IM):** Injected into muscle — used for IGF-1 LR3 (local muscle effect), some GHRP protocols
- **Intranasal:** Nasal spray — Semax, Selank, DSIP can be administered this way (less efficient but needlefree)
- **Oral:** Most peptides are degraded by digestive enzymes; generally ineffective. Exception: BPC-157 has oral activity for gut conditions

■ 2. BPC-157 — BODY PROTECTIVE COMPOUND

Full name: Body Protection Compound-157. A 15-amino acid peptide derived from a protective protein found in human gastric juice. BPC-157 is the most versatile peptide available, with healing effects documented across muscles, tendons, ligaments, gut, nerves, and brain.

Mechanism of Action:

- **Angiogenesis:** Upregulates VEGF (Vascular Endothelial Growth Factor) — grows new blood vessels to injured tissue, dramatically accelerating healing
- **Growth factor modulation:** Increases expression of PDGF, EGF, and bFGF — key signaling proteins for tissue repair
- **Nitric oxide pathway:** Modulates NO production; vasodilatory effects improve local blood flow to injury site
- **Gut healing:** Repairs intestinal epithelium; increases gastric mucosal blood flow; protects against NSAID-induced damage
- **Tendon and ligament repair:** Accelerates collagen organization and tendon fibroblast proliferation by 2–3×

Application	Dose	Protocol	Route	Duration
Injury healing (tendon/muscle)	250–500 mcg/day	Split into 2 injections	SC near injury site	Until healed (typically 4–8 wks)
Gut healing (IBS, ulcers, leaky gut)	250 mcg 2×/day	Morning + evening	SC or oral (capsule)	4–8 weeks
Systemic recovery & prevention	200–300 mcg/day	Once daily	SC (abdominal)	Cycle 8 wks on / 4 wks off
Brain/nerve healing	200–400 mcg/day	2× daily	SC (nape of neck area or systemic)	4–6 weeks

> ■ *Animal studies (Sikiric et al., multiple publications in Journal of Physiology-Paris) demonstrate BPC-157 heals Achilles tendon transection 2–3× faster than untreated controls, with superior collagen organization in healed tissue. Human trials are currently in Phase II for IBD.*

■ 3. TB-500 (THYMOSIN BETA-4)

TB-500 is a synthetic analog of Thymosin Beta-4 (Tβ4), a naturally occurring 43-amino acid protein found in virtually all human cells. It regulates **actin polymerization** — the process that drives cellular movement, wound healing, and tissue repair throughout the body.

- **Mechanism:** Sequesters G-actin monomers, promoting cell migration to injury sites; activates wound healing cascades; potent anti-inflammatory via NF-κB pathway inhibition
- **Systemic effect:** Unlike BPC-157 (local), TB-500 acts systemically — benefits whole-body healing simultaneously

- **Cardiovascular:** Promotes cardiac cell survival after injury; studied in heart failure patients (Phase II trials)
- **CNS healing:** Promotes neural stem cell migration; studied for stroke recovery

Phase	Dose	Frequency	Duration
Loading (active injury)	2–2.5 mg	2× weekly	6 weeks
Maintenance (prevention)	2 mg	Once per week	Ongoing (cycle 12 wks on / 4 wks off)
Performance (recovery)	1–2 mg	Once weekly	8–12 weeks

BPC-157 + TB-500 Stack (Gold Standard Healing Protocol):

- BPC-157 250 mcg + TB-500 2 mg on the same day, both SC — synergistic local + systemic healing
- Often called 'Wolverine Stack' — dramatically accelerates recovery from almost any musculoskeletal injury
- Can safely be run alongside any other protocol (AAS, SARMs, natural)

4–7. GHK-Cu · SEMAX · SELANK · EPITHALON

GHK-Cu (COPPER PEPTIDE)

A naturally occurring tripeptide (Gly-His-Lys) that binds copper. Found in blood, saliva, and urine; declines with age.

Mechanism: Activates ~4,000 human genes; upregulates collagen, elastin, and antioxidant enzymes; anti-inflammatory; promotes hair follicle survival; potent wound healing

Protocol: Skin injection/topical: 0.1–2mg/day SC or applied topically as 0.1–1% cream. Best for skin aging, hair loss prevention, wound repair.

SEMAX (ACTH 4-7-Pro-Gly-Pro)

Synthetic analog of the neuropeptide ACTH. Developed in Russia in the 1980s for cognitive enhancement and stroke rehabilitation.

Mechanism: Increases BDNF (Brain-Derived Neurotrophic Factor) by up to 800%; enhances dopamine and serotonin signaling; potent neuroprotective effects; improves memory consolidation

Protocol: 600–900 mcg/day intranasal (3 drops each nostril, 2–3 daily). Use in 2–4 week cycles for cognitive enhancement. Excellent for study, work performance, or post-injury neurological recovery.

SELANK (TKP-OH)

Synthetic heptapeptide analog of Tuftsin. Anxiolytic and nootropic — developed at the Russian Institute of Molecular Genetics.

Mechanism: Modulates GABAergic system (similar mechanism to benzodiazepines but without tolerance/dependence); increases serotonin metabolism; anti-anxiety without sedation; improves information processing speed

Protocol: 250–500 mcg/day intranasal. Use for: anxiety disorders, social anxiety, cognitive stress, PCT mood management (excellent for hormonal transitions).

EPITHALON (EPITALON)

Tetrapeptide (Ala-Glu-Asp-Gly) synthesized by the pineal gland. The most studied anti-aging peptide in existence.

Mechanism: Activates telomerase enzyme — extends telomere length in aging cells; normalizes melatonin production; antioxidant; tumor suppressive in animal models; extends lifespan in animal studies by 24–68%

Protocol: 5–10 mg/day SC for 10-day cycles, 2–3 times per year. Often combined with sleep protocols. Intranasal option: 20 mcg/day.

8–10. PT-141 · AOD-9604 · DSIP

PT-141 (BREMELANOTIDE)

Melanocortin receptor agonist (MC3R, MC4R) — works centrally in the brain's limbic system (hypothalamus) to generate sexual arousal. The only peptide that directly activates desire rather than addressing peripheral circulation (unlike sildenafil/Viagra).

Protocol: 1–2 mg SC 45–90 min before sexual activity. Side effects: temporary nausea (use 0.5 mg to test), flushing. Cycle 2×/week maximum. Approved by FDA (2019) as Vyleesi for women's hypoactive sexual desire disorder.

AOD-9604

The C-terminal fragment of human growth hormone (HGH amino acids 176–191) that specifically targets fat metabolism without the growth-promoting or insulin-dysregulating effects of full HGH.

Protocol: 500 mcg/day SC, fasted AM injection. Directly stimulates β3-adrenergic receptors in fat cells → lipolysis. No effect on blood glucose or IGF-1. Ideal addition to cutting protocols for targeted fat loss.

DSIP (DELTA SLEEP-INDUCING PEPTIDE)

Naturally occurring neuropeptide in brain and GI tract. Reduces ACTH and cortisol; modulates sleep architecture by promoting deep slow-wave (delta) sleep. Also protects against oxidative stress.

Protocol: 100–200 mcg SC 30–60 min before bed. Use in 2-week cycles. Excellent for athletes with high stress loads, post-training insomnia, or those needing improved recovery sleep quality.

11. STRATEGIC STACKING GUIDE

Goal	Stack	Daily Protocol	Duration
Injury recovery & performance	BPC-157 + TB-500	BPC-157 250mcg AM + PM; TB-500 2mg 2×/week	6–8 weeks

Goal	Stack	Daily Protocol	Duration
Mass building (with AAS/SARMs)	IGF-1 LR3 + GHRP-6 + CJC-1295	IGF-1 LR3 50mcg post-workout; GHRP-6 100mcg 3×/day; CJC-1295 DAC 2mg/wk	12–16 weeks
Anti-aging & longevity	Epithalon + GHK-Cu + BPC-157	10-day Epithalon cycle 2×/year; GHK-Cu daily; BPC-157 ongoing low dose	Year-round
Cognitive optimization	Semax + Selank	Semax 600mcg AM intranasal; Selank 250mcg PM intranasal	2–4 week cycles
Fat loss acceleration	AOD-9604 + HGH Fragment 176-191	AOD 500mcg fasted AM; HGH Fragment 500mcg split AM/pre-workout	12 weeks
Sleep & recovery	DSIP + BPC-157 + Epithalon	DSIP 150mcg pre-bed; BPC-157 250mcg AM; Epithalon 10-day cycles	Cyclical

■ 12. RECONSTITUTION, STORAGE & INJECTION GUIDE

Reconstitution Protocol:

- Use only bacteriostatic water (BAC water) — not sterile water. BAC water contains 0.9% benzyl alcohol which prevents microbial growth
- Inject BAC water slowly down the SIDE of the vial — never directly onto the peptide powder (can degrade it)
- Swirl gently to mix — never shake vigorously (breaks peptide bonds)
- Standard reconstitution: 2mL BAC water per 5mg vial = 2.5 mcg/µL concentration
- Use 1mL insulin syringe (U-100) for all SC injections — smallest gauge available (27–31G)

Storage Rules:

- Lyophilized (dry powder): Stable at room temperature for 6–12 months, or refrigerated for 2+ years
- Reconstituted solution: Refrigerate at 2–8°C; stable for 4–6 weeks; discard if cloudy or precipitate forms
- Never freeze reconstituted peptides (ice crystals destroy structure)
- Protect all peptides from light — UV radiation degrades amino acid bonds

■■ **IMPORTANT:** Source peptides only from verified, reputable suppliers with third-party HPLC purity testing certificates. Impure peptides are at best ineffective and at worst dangerous. Always request a Certificate of Analysis (CoA) before purchasing.